Sungyeon Kook

kooksungyeon310@gmail.com · Website · Google Scholar · ORCID · LinkedIn

Research Interests

Quantum Error Correction, Fault-Tolerant Quantum Computing, Quantum Information Theory, Quantum Algorithms, Error-Correcting Codes

Education

Korea University — Seoul, Korea
M.S. (Master of Science) in Electrical Engineering — Mar. 2024 – Feb. 2026

Korea University — Seoul, Korea
B.S. (Bachelor of Science) in Electrical Engineering — Mar. 2020 – Feb. 2024

Research Experience

Korea Institute of Science and Technology Information (KISTI) — Daejeon, Korea
Post-Master Researcher, Quantum Network Research Center (Advisor: Dr. Ilkwon Sohn) — Mar. 2026 – Present

Korea University — Seoul, Korea
Graduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo) — Mar. 2024 – Feb. 2026

Korea Research Institute of Standards and Science (KRISS) — Daejeon, Korea
Visiting Student Researcher, Center for Superconducting Quantum Computing System (Advisor: Dr. Hwan-Seop Yeo) — Aug. 2024 – Aug. 2025

Korea University — Seoul, Korea
Undergraduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo) — Jan. 2023 – Feb. 2024
Undergraduate Research Assistant, Quantum Control Laboratory (Advisor: Prof. Mahn-Soo Choi) — Jul. 2023 – Aug. 2023

Publications

1. Changyeol Lee†, **Sungyeon Kook**†, Wooyeong Song, Kwangil Bae, and Ilkwon Sohn, "Quantum Crandall reduction for pseudo-Mersenne moduli: structure-aware circuits and surface-code resource analysis," manuscript in preparation for submission to *Quantum Science and Technology*. († Equal contribution.)

2. Wooyeong Song, **Sungyeon Kook**, Wonhyuk Lee, and Ilkwon Sohn, "Asymmetry-aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client," under review at *IEEE Network*. arXiv:2607.11238.

3. **Sungyeon Kook**, Yujin Kang, Ilkwon Sohn, and Jun Heo, "Low Spatial Cost CCZ Magic State Factory," under review at *Quantum Information Processing*. arXiv:2606.24170.

4. **Sungyeon Kook**, Yujin Kang, and Jun Heo, "Limited Connectivity Quantum Architectures: Enhancing Surface Code Implementation Through Routing Qubits," in *Proceedings of the 15th International Conference on Information and Communication Technology Convergence (ICTC)*, 2024. DOI: 10.1109/ICTC63764.2024.10826938.

5. Yujin Kang, Youshin Chung, Huidan Zheng, **Sungyeon Kook**, and Jun Heo, "Magic State Distillation with Reduced Time Cost," in *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024. DOI: 10.1109/QCE60285.2024.00142.

Conferences

Poster Presenter
- 2026 Annual Meeting of the Quantum Information Society of Korea — 2026.02
- International Conference on Information and Communication Technology Convergence (ICTC) — 2025.10
- **Quantum Error Correction Conference (QEC 2025)** — 2025.08

Oral Presenter
- Joint Conference on Communications and Information (JCCI) — 2025.04
- International Conference on Information and Communication Technology Convergence (ICTC) — 2024.10
- KICS Conferences (Multiple Oral Presentations) — 2023–2025

Co-author
- **Asian Quantum Information Science Conference (AQIS 2026)** — 2026.08
- Optics and Photonics Congress 2026 (OPC 2026) — 2026.07
- **IEEE International Conference on Quantum Computing and Engineering (QCE)** — 2024.09

Research Projects

** Journal Published or under review*

Low-Spatial-Cost CCZ Magic-State Factory* — *2025–Present*

- Converted the $[[8,3,2]]$ CCZ distillation circuit into **eight Z-basis $\pi/8$ Pauli-product joint measurements**, reducing the factory from **12 to 4 persistent logical patches**.
- Eliminated **eight redundant $|0\rangle$ ancilla qubits** while preserving logical functionality and single-fault detection.
- Designed a surface-code CCZ factory using **four-qubit twisted ZY measurements**, reducing spatial cost by **71.4%** and space-time volume by **37.7%**.

Surface-Code Decoder and Code Switching* — *2024–Present*

- Built a **Stim–PyMatching decoding pipeline** for rotated surface codes, evaluating logical failure rates across code distances and physical-error rates.
- Developed an **erasure-aware MWPM decoder** by dynamically reweighting matching-graph edges using known erasure locations.
- Proposed a fault-tolerant **Bacon–Shor-to-surface-code switching protocol** for heterogeneous quantum computation.

Architecture-Aware Quantum Circuit Mapping and Scheduling — *2024–2025*

- Designed Heavy-Hex surface-code mapping strategies to minimize SWAP overhead while balancing flag-qubit and hardware resources.
- Developed a **QODG-based scheduling framework** for checkerboard architectures by modeling gate dependencies, resource conflicts, and conditional execution latency.
- Implemented a Greedy DAG Qiskit transpiler pass, reducing average circuit depth from **73 to 61.4**.

Fault-Tolerant Resource Analysis for Quantum Modular Reduction* — *2025–Present*

- Developed a **surface-code resource-estimation framework** incorporating code distance, logical-error requirements, magic-state factories, decoder throughput, and backlog.
- Formulated a **fault-tolerant runtime model** incorporating T-depth, feed-forward latency, and decoder performance.
- Reduced estimated execution time from **53.77 ms to 30.05 ms** under Sparse Blossom decoding.

Experimental Superconducting Quantum Computing — *KRISS, 2024–2025*

- Developed Python-based experimental-control software for superconducting-qubit experiments using Zurich Instruments hardware.
- Performed **qubit calibration and characterization**, including spectroscopy, randomized benchmarking, DRAG calibration, and cross-resonance gates.
- Investigated Heavy-Hex architectures qubit routing, flag-qubit syndrome extraction, and connectivity-aware fault-tolerant circuits.

Additional Research

Variational Quantum Algorithms — *2023*

- Implemented **VQE, QAOA, and quantum-kernel algorithms** using Qiskit and Xanadu PennyLane.

Resource-Efficient Data-Bus Architectures for Fault-Tolerant Quantum Computing — *2024*

- Co-authored a QEC conference paper and contributed to decoder implementation, Stim simulation, and fault-tolerant code switching.

Technical Skills

Programming: Python, C++, MATLAB

Quantum Software: Qiskit, Pennylane, Stim, PyMatching

Experimental Platforms: Zurich Instruments, Superconducting Quantum Computing

Academic Service

QUantum Information Club at Korea university (QUICK)

Founder and Treasurer — Jan. 2023 – Dec. 2023

- Co-founded the first undergraduate quantum information club at Korea University.
- Managed club administration and event organization.
- Organized the first IBM Qiskit Fall Festival hosted at Korea University.

Teaching & Honors

Outstanding Academic Achievement, Korea University — Spring 2023

Teaching Assistant, Data Science and Artificial Intelligence — Fall 2024

Provided programming and data analysis instruction for undergraduate students.

Teaching Assistant Scholarship, Korea University — Fall 2024